Integrating f'(x)/f(x)

Exam code: 9MA0

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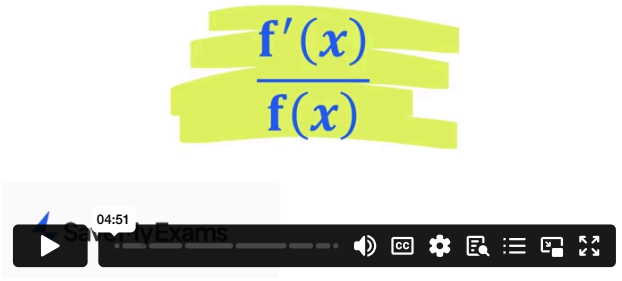
Updated on 23 June 2025



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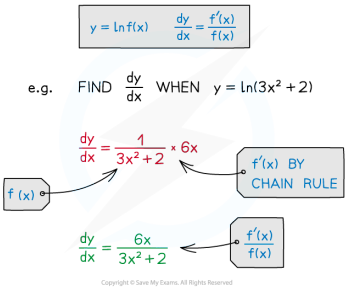
Help me understand

Integrating f'(x)/f(x)

How do I integrate 1/x?

- The technique for integrating fractions depends on the type of fraction
- For **polynomial denominators** see Integration using Partial Fractions
- If $\frac{dy}{dx} = \frac{1}{x}$ then **y = ln |x| + c** – see Integrating Other Functions
- The type of fraction dealt with here is a specific case of Reverse Chain Rule



How do I integrate f'(x)/f(x)?

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

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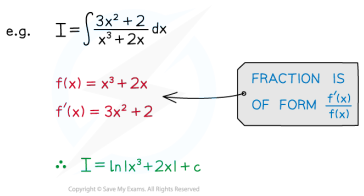
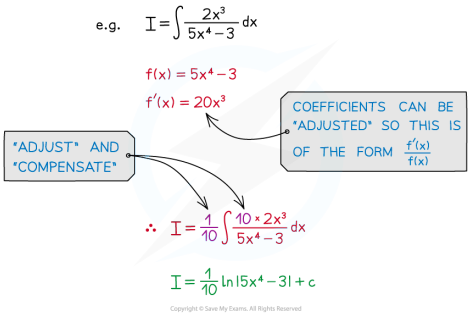
- "The top is 'almost' the derivative of the bottom"
  - 'almost' here meaning 'a multiple of' (see below)
- The integral will involve **ln |f(x)|** – ie **ln** of the bottom
  - Due to reverse chain rule



How do I adjust a fraction into the form f'(x)/f(x)?



- There may be coefficients to 'adjust' and 'compensate' for

Examiner Tips and Tricks

- If you're unsure if the fraction is of the form **f'(x)/f(x)**, differentiate the denominator.
- Compare this to the numerator but you can ignore any **coefficients**.
- If the coefficients do not match then 'adjust' and 'compensate' for them.

Worked Example

$$I = \int_{\frac{\pi}{12}}^{\frac{\pi}{2}} \frac{\sin 4x}{1 + 2\cos 4x}\,dx$$

$f(x) = 1 + 2\cos 4x$

$f'(x) = 2 \times -\sin 4x \times 4$

$= -8\sin 4x$

"ADJUST" AND "COMPENSATE"

IGNORING COEFFICIENTS, I IS OF THE FORM $\frac{f'(x)}{f(x)}$

$$I = -\frac{1}{8}\int_{\frac{\pi}{12}}^{\frac{\pi}{2}} \frac{-8\sin 4x}{1 + 2\cos 4x}\,dx$$

$$I = \left[-\frac{1}{8}\ln|1 + 2\cos 4x|\right]_{\frac{\pi}{12}}^{\frac{\pi}{2}}$$

"ln OF BOTTOM"

$$I = \left(-\frac{1}{8}\ln|1 + 2\cos 2\pi|\right) - \left(-\frac{1}{8}\ln\left|1 + 2\cos\frac{\pi}{3}\right|\right)$$

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$$I = \left(-\frac{1}{8}\ln 3\right) - \left(-\frac{1}{8}\ln 2\right)$$

$$I = \frac{1}{8}(\ln 2 - \ln 3)$$

SWAPPED TERMS SO FIRST IS POSITIVE

$$\therefore\ I = \frac{1}{8}\ln\frac{2}{3}$$

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